Hyper-Temporal Data Based Modulation Transfer Functions Compensation for Geostationary Remote Sensing Satellites

Xue Yang, Liang Liang, Feng Li, *Senior Member, IEEE*, Qingjiu Tian, Xiaotian Lu, Lei Xin, Yi Guo, and Wenjun Dong

***Abstract*— Over the past years, the acquisition of hyper-temporal data (HTD) from geostationary orbit remote sensing satellites (GEORSS) has provided numerous new research opportunities. Many factors influence the in-orbit dynamic modulation transfer function (MTF) of GEORSS, making it difficult to satisfy the requirements for space-borne cameras. The MTF compensation (MTFC) technique can effectively optimize the design of dynamic MTFs for GEORSS. The traditional MTFC methods mainly consider the sensor, atmosphere and relative motion of the satellite platform when improving GEORSS image quality. They will introduce new high frequency noise, resulting in image information loss. In this article, a mixed sparse higher-order non-convex total variation (MS-HONCTV) model-aided MTFC method is proposed. By introducing the group sparse regularization (GSR) term into the MS-HONCTV model, it increases the robustness to noise and hence reduces the degeneration of the MTF produced by satellite' low pointing stability. The MS-HONCTV model is then applied to solve the problem of image degradation. The quality of remote sensing (RS) data is improved by the proposed MTFC and this is achieved without modifying the aperture diameter, focal length, or detector size of the satellite's optical system. Experimental results show that the proposed MS-HONCTV effectively improves the images' MTF, SNR, gray mean gradient (GMG), and standard deviation (SD), as evidenced by subjective qualitative analysis and objective quantitative assessments of simulated data, laboratory data, and GF-4 satellite data. Compared with other methods, the SNR of the proposed method is increased by 30%, GMG by 14.21%, and SD by 6.3% on average.**

***Index Terms*— Geostationary orbit remote sensing satellites (GEORSS), GF-4, hyper-temporal data (HTD), mixed sparse higher-order nonconvex total variation (MS-HONCTV), modulation transfer function compensation (MTFC).**

I. Introduction

THE concept of hyper-temporal imaging (HTI) was first proposed by Srivastava and Kumar [1]. Hyper-temporal data (HTD) would ideally have high temporal and spatial resolution and is more advantageous in continuous monitoring of key areas, image classification and target recognition. Each frame of data contains unique high-frequency information because space cameras are detector limitation systems suffering under-sampling problems. By collecting these high-frequency components in each frame, MTF compensation (MTFC) methods can be used to reconstruct high-resolution (HR) images [2]. The acquisition of HTD from geostationary orbit remote sensing satellites (GEORSS) has now been receiving increasing attention.

An optical remote sensing (RS) satellite in geostationary orbit has the natural advantage of being able to capture a series of images from the same region over a very short time. GEORSS are now producing an unprecedented amount of HTD owing to the continuous efforts in developing a new generation of GEORSS [3]. For example, the frame rate of HTD for the GF-4 satellite is 60 frame/h, the frame rate of HTD for FY-4 satellite is four frame/h [4], [5]. A spaceborne camera of a GEORSS usually has large aperture lenses, long focal length, and low MTF. Many factors, such as satellite attitude stability, atmospheric disturbance, and detector sampling, among others, cause image degradation and compromise image quality [6], [7]. Because of the limitations of optical imaging systems (e.g., volume, weight, and development cost), solely improving hardware is not a viable solution to improve the quality of the images from RS satellites [8].

MTFC is necessary and widely used by numerous satellite companies to enhance the in-orbit instruments' dynamic MTF. Studies show that the image quality of optical remote sensors with the same spatial resolution after MTFC is better than that of optical remote sensors with relatively large apertures [9]. MTFC improves the MTF of the optical imaging system

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by recovering the high-frequency information lost in the acquisition, transmission, and processing of the optical image, so as to obtain sharp edges and richer details of the images.

The existing MTFC methods of RS images mainly adopt blind image deconvolution (BID) and nonblind image deconvolution (NBID). Most of the datasets processed by these methods are single-frame RS images. The NBID method restores the LR image from the blur kernel obtained in advance. NBID methods have low computational complexity and high restoration quality, such as Richardson–Lucy (RL) Algorithm [10], Wiener filtering (WF) [11], and hyper-Laplacian Priors (HLP) [12] and so on. Because of the complex structure of RS images, there is, however, unknown prior information, so it is difficult to apply the NBID method in the MTFC of RS images. In BID, we need to recover the LR image from an unknown blur kernel. There are many approaches of this kind, e.g., the Tikhonov regularization (TR) method [13], Total variation (TV) regularization [14], the local minimal intensity prior (LMIP) image deblurring method [15], and so on. The BID method helps to alleviate image quality degradation in RS images; however, the method has high computational complexity and low restoration accuracy. Several studies have been carried out on the BID method. Kaur et al. [16] developed a fast BID method based on Split Bregman (SB) to improve calculation speed. Liu et al. [17] used the higher-order nonconvex total variation model (HONCTVM) to achieve image noise removal, but the high gradient ignores the weak texture and detail information of the image, resulting in blurred local detail information of the image. Adam et al. [18] introduced sparsity into the high-order total variational model (HOTVM) to achieve image blur removal by exploiting self-similarity within images. Although this method can avoid staircase artifacts well, it cannot recover the edge details effectively. We summarize a detailed comparison as shown in Table I, where the applicable data sets, major contributions, SNR improvement, and shortcomings are included.

As shown in the above table, we can see that the existing total variation models have achieved good results in image denoising and deblurring with the cost of losing detailed information. Most of these methods are, however, based on a single temporal image. They were rarely applied in MTF compensation for multitemporal images. In order to make full use of specific information of each frame in multitemporal images, we propose a mixed sparse MS-HONCTV model to implement MTFC, eliminating staircase artifacts for GEORS. MS-HONCTV model introduces group sparse regularization (GSR) and nonconvex second-order regularization into the total variation model to improve image sharpness, signal-to-noise ratio (SNR), and MTF quality.

The primary contributions of this article are summarized as follows.

1) A theoretical validation is conducted for the HTI technology with MTFC calculation methods to improve the MTF and image quality of GEORS.
2) The GSR technique is introduced into the HONCTV model, and a HONCTV model with mixed sparse representation is established. The iterative re-weighted ℓ_1 (IRL_1) method and the majorization minimization (MM) algorithm are introduced into the alternating direction method of multipliers (ADMM) in order to obtain the optimal solution of the model.
3) Extensive experiments have been conducted, demonstrating that the MS-HONCTV aided MTFC method eliminates staircase artifacts, recovers more structure and texture, shows better performance, and effectively improves the image MTF, SNR, gray mean gradient (GMG), and standard deviation (SD).

TABLE I
COMPARISON OF EXISTING MTFC METHODS

Method	Data Type	Major Contributions	Short Comings
RL [10]	Single temporal RS image	Strengthen the image edges	Low restoration accuracy
WF [11]	Single temporal Saturn image	Removal image noise	Restoration accuracy is limited
HLP [12]	Single temporal nature image	Image deblurring	Low efficiency overlapping area artifacts
TR [13]	Single temporal satellite image	Suppress noise and restore details	Result local details are blurred
TV [14]	Single temporal Lena image	Preserve sharp edges of images	Produce staircase effects
LMIP [15]	Single temporal RS image	Enhance the image edge information	Produce artifact effects
SB [16]	Single temporal RS image	suppress the ringing effect	Local information is lost
Liu [17]	Single frame nature image	Reduce staircase effects	Local details of the image are smooth
Adam [18]	Single frame nature image	Eliminate staircase effects	Restore local details are not obvious

The rest of this article is organized as follows. Section II describes MTF numerical simulation. Section III-A introduces modeling a degrading image MTF, and Section III-B establishes MTFC based on the MS-HONCTV model. Section IV presents experimental results and analysis. Finally, Section V concludes the article.

II. NUMERICAL ANALYSIS

Factors such as optical lenses, detectors, and satellite platforms interact to degrade the MTF of spaceborne cameras. The impact of the MTF on the imaging system may be the product of the MTFs of the optical system components of the satellite. The combined action of detector pixel integral imaging and the detector array discrete sampling produces the MTF of the RS detector. Suppose the filling factor is not 100%, then the overall MTF of the detector is the product of the pixel and sampling MTFs [19], [20], [21]

$$\mathrm{MTF}_{\mathrm{det}}(\upsilon) = \mathrm{MTF}(\upsilon)_{\mathrm{pixel}} \times \mathrm{MTF}(\upsilon)_{\mathrm{sampling}} \tag{1}$$

where υ is the input signal's spatial frequency.

Two frames of images with pixel displacement are acquired using the same array CMOS detector to capture the same

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region. Assume that two frames of images are staggered by half the pixel spacing in the vertical and horizontal directions, and the information between the images is complimentary. At this point, the sampling interval becomes $p/2$, and the system's Nyquist frequency shifts to $1/p$. The detector's MTF can be expressed as

$$\mathrm{MTF}_{\mathrm{det}} = \mathrm{sinc}(dv) \times \mathrm{sinc}\left(\frac{pv}{2}\right). \tag{2}$$

During the HTD-imaging process, n LR images can be obtained in the same field of view; we assume pixel displacement is $1/n(n > 0)$, the sampling interval becomes $1/n$, and the Nyquist frequency ideally increases to n times that of the original single-frame LR image. The MTF of the system takes the form

$$\mathrm{MTF}_{\mathrm{det}} = \mathrm{sinc}(dv) \times \mathrm{sinc}\left(\frac{pv}{n}\right). \tag{3}$$

According to (2) and (3), if the size of the RS detector remains constant, the MTF output grows as the sampling frequency increases, and the image quality can be improved accordingly. This implies that the hyper-temporal images-based MTFC technique is valid in theory, although those HTD images are obtained with random rather than regular displacement. The HTD acquired with GEORSS has random sub-pixel displacements between consecutive frames. The detailed information between image frames is different and complementary to each other. Compared with the general spaceborne detector imaging method, this imaging method has the advantage of ensuring that the pixel integration has sufficient luminous flux and a higher spatial sampling frequency. When the pixel size is the same, the sampling frequency is doubled, and the spatial resolution can also be improved. Therefore, the theory based on MTFC technology in improving HTD is correct.

III. Methodology

A. Modeling a Degrading Procedure of MTF

Establishing an accurate MTF degradation model is the premise for MTFC. Assume that an ideal HR image is degraded into a sequence of LR satellite images after a series of atmospheric disturbances, object motion, defocusing, and noise interference. These factors can be combined in the following mathematical model:

$$\boldsymbol{g}_i = B_i M_i \boldsymbol{z} + \boldsymbol{n}_i, \quad i = 1, 2, \ldots, N \tag{4}$$

where B_i refers to the blur matrix of the serial frame satellite image caused by the platform vibration, atmospheric scattering and absorption, and other radiative distortions that occur in the process of satellite image acquisition. M_i is the image-warping matrix corresponding to possible warping effect. $\boldsymbol{z}$ represents the HR satellite image to be recovered. $\boldsymbol{n}_i$ is the zero-mean Gaussian noise for any interference and contamination. N is the total number of observed LR images.

According to the degradation model (4), the MTFC process for multiframe-sequence images includes the blur matrix B_i and warping matrix M_i of each frame image of the sequence $\boldsymbol{g}_i$. The blur matrix B_i is a circular matrix and can be approximated by a Gaussian blur kernel. The high-precision image-warping matrix M_i between the images is calculated by using an image registration method based on the Oriented FAST and Rotated BRIEF (ORB) algorithm and a combination of gray registration [3]. In this process, the displacement difference between images after registration reaches the subpixel level, and the calculated MTFC image quality exceeds that of the conventional MTFC methods.

The MTFC of the HTD exploits the nonredundant complementary information existing in each LR-image frame to recover the high-frequency components that is overlapped in the image. This allows for effective compensation of the image MTF for amplitude-frequency characteristics. Since the process of RS image acquisition is irreversible, (4) describes the inverse of the imaging process. The problem of compensating for the MTF of the HTD is the inverse calculation of the degraded image process.

B. MTFC Based on Mixed Sparse HONCTV Model

Consider that (4) describes a problem that is ill-conditioned; the prior knowledge of the HR image $\boldsymbol{z}$ has to be introduced as additional information to regularize the problem and eliminate trivial solutions. One of the most successful regularization methods is the HONCTV regularizer [22]. This is due to its ability to preserve sharp edges in images, attenuating the ring artifact in the image appropriately smooth area, which is important for image compensation. The HONCTV method, however, directly ignores the texture and detail information of the image, resulting in blurred detail information of the image.

Mixed sparse technique combined with the HONCTV method can effectively express image sparsity and smoothness characteristics, which can not only reveal image details but also avoid image local smoothness. The expression of the MS-HONCTV model is as follows:

$$\min_{z} \frac{\lambda}{2}\left|\left|B_i M_i \boldsymbol{z} - \boldsymbol{g}_i\right|\right|_2^2 + \phi(\nabla \boldsymbol{z}) + \omega\left|\left|\nabla^2 \boldsymbol{z}\right|\right|_p^p \tag{5}$$

where $\phi(\cdot)$ and $||\cdot||_p^p$ correspond to the group sparse regularizer and nonconvex ℓ_p norm $(0 < p \leq 1)$ regularizers, respectively; λ and ω are nonnegative regularization parameters that control data fidelity and higher-order nonconvex regularization, respectively; $\nabla \boldsymbol{z}$ and $\nabla^2 \boldsymbol{z}$ are the first- and second-order gradient operators, respectively, assuming periodic boundary conditions for the RS image.

In order to solve (5), the ADMM method [23] is employed after converting to the following separable optimization problem by introducing two auxiliary variables:

$$\begin{aligned} &\min_{z} \ \frac{\lambda}{2}\left|\left|B_i M_i \boldsymbol{z} - \boldsymbol{g}_i\right|\right|_2^2 + \phi(\boldsymbol{v}) + \omega||\boldsymbol{w}||_p^p \\ &\text{s.t.} \quad \boldsymbol{v} = \nabla \boldsymbol{z}, \quad \boldsymbol{w} = \nabla^2 \boldsymbol{z}. \end{aligned} \tag{6}$$

It is further transformed to an unconstrained problem by using augmented Lagrangian as follows:

$$\begin{aligned} &L_{\mathcal{A}}(\boldsymbol{z}, \boldsymbol{v}, \boldsymbol{w}; \mu_1, \mu_2) \\ &\quad = \frac{\lambda}{2}\left|\left|B_i M_i \boldsymbol{z} - \boldsymbol{g}_i\right|\right|_2^2 + \phi(\boldsymbol{v}) + \omega||\boldsymbol{w}||_p^p - \mu_1^{\mathrm{T}}(\boldsymbol{v} - \nabla \boldsymbol{z}) \\ &\qquad + \frac{\rho}{2}||\boldsymbol{v} - \nabla \boldsymbol{z}||_2^2 - \mu_2^{\mathrm{T}}\left(\boldsymbol{w} - \nabla^2 \boldsymbol{z}\right) + \frac{\rho}{2}\left|\left|\boldsymbol{w} - \nabla^2 \boldsymbol{z}\right|\right|_2^2 \end{aligned} \tag{7}$$

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where μ_1 and μ_2 are the Lagrange multipliers, $\rho(\rho > 0)$ is the approximal parameter, and its value is related to $||\boldsymbol{v} - \nabla \boldsymbol{z}||_2^2$ and $||\boldsymbol{w} - \nabla^2 \boldsymbol{z}||_2^2$.

The optimization strategy for solving each subproblem of $\boldsymbol{z}$, $\boldsymbol{v}$, $\boldsymbol{w}$ is the following while holding others fixed.

1) Solving $\boldsymbol{z}$ is a least-squares problem

$$\begin{aligned}\boldsymbol{z}^{k+1} = \underset{\boldsymbol{z}}{\operatorname{argmin}} \frac{\lambda}{2}\left|\left|B_i M_i \boldsymbol{z} - \boldsymbol{g}_i\right|\right|_2^2 - \mu_1^{\mathrm{T}}\left(\boldsymbol{v}^k - \nabla \boldsymbol{z}\right) \\ + \frac{\rho}{2}\left|\left|\boldsymbol{v}^k - \nabla \boldsymbol{z}\right|\right|_2^2 - \mu_2^{\mathrm{T}}\left(\boldsymbol{w}^k - \nabla^2 \boldsymbol{z}\right) \\ + \frac{\rho}{2}\left|\left|\boldsymbol{w}^k - \nabla^2 \boldsymbol{z}\right|\right|_2^2. \end{aligned} \tag{8}$$

This $\mathbf{z}$-subproblem has a closed-form solution as

$$\begin{aligned}&\boldsymbol{z}^{k+1} \\ &= \left[\lambda (B_i M_i)^{\mathrm{T}} (B_i M_i) + \rho \nabla^{\mathrm{T}} \nabla + \rho \left(\nabla^2\right)^{\mathrm{T}} \nabla^2 + \rho \mathbf{I}\right]^{-1} \\ &\quad \cdot \left(\lambda (B_i M_i)^{\mathrm{T}} \boldsymbol{g}_i - \nabla^{\mathrm{T}} \mu_1 + \rho \nabla^{\mathrm{T}} \boldsymbol{v} - \left(\nabla^2\right)^{\mathrm{T}} \mu_2 + \rho \left(\nabla^2\right)^{\mathrm{T}} \boldsymbol{w}\right)\end{aligned} \tag{9}$$

where $\boldsymbol{I}$ is the identity matrix. Assume that $\boldsymbol{z}$ satisfies the periodic boundary condition. The matrices $(B_i M_i)^{\mathrm{T}}(B_i M_i)$, $\nabla^{\mathrm{T}} \nabla$, and $(\nabla^2)^{\mathrm{T}} \nabla^2$ are block circulant matrices, and these matrices can be diagonalized by a 2-D fast Fourier transform (FFT).

2) The v-subproblem has the following form:

$$\begin{aligned}\boldsymbol{v}^{k+1} &= \underset{\boldsymbol{v}}{\operatorname{argmin}} \frac{\rho}{2}\left|\left|\boldsymbol{v} - \nabla \boldsymbol{z}^{k+1}\right|\right|_2^2 - \mu_1^{\mathrm{T}}\left(\boldsymbol{v} - \nabla \boldsymbol{z}^{k+1}\right) + \phi(\boldsymbol{v}) \\ &= \underset{\boldsymbol{v}}{\operatorname{argmin}} \frac{\rho}{2}\left|\left|\boldsymbol{v} - \left(\nabla \boldsymbol{z}^{k+1} + \frac{\mu_1^k}{\rho}\right)\right|\right|_2^2 + \phi(\boldsymbol{v}).\end{aligned} \tag{10}$$

This is an overlapping group sparse total variation (OGSTV) denoising problem [23]. Since the MM algorithm has the advantage of efficiently using the convexity of functions to search for the maximum and minimum values of the original function, when the optimization of the original objective function is difficult, an alternative function that is easy to optimize and close to the original objective function is solved [24]. Therefore, this article uses a quadratic function to replace $\phi(\boldsymbol{v})$ to realize the MM algorithm

$$\phi(\boldsymbol{v}) = \sum\nolimits_l \left[\sum\nolimits_{j=0}^{K-1} |\boldsymbol{v}(l+j)|^2\right]^{\frac{1}{2}} \tag{11}$$

where K is the size of the group, and $(l,\ j)$ is the corresponding pixel on image $\boldsymbol{g}_i$. The group size can be seen as a K-sized window starting with index l. In each iteration of the MM algorithm, a convex quadratic problem is minimized, yielding a convergent solution to (11).

3) In the same way, the $\boldsymbol{w}$-subproblem is as follows:

$$\boldsymbol{w}^{k+1} = \underset{\boldsymbol{w}}{\operatorname{argmin}} \frac{\rho}{2}\left|\left|\boldsymbol{w} - \left(\nabla^2 \boldsymbol{z}^{k+1} + \frac{\mu_2^k}{\rho}\right)\right|\right|_2^2 + \omega ||\boldsymbol{w}||_p^p. \tag{12}$$

Equation (12) is a nonconvex optimization problem. Nonconvex optimization algorithm and threshold selection determine the image denoising effect. We use the IRL_1 algorithm proposed by [25] to balance the positive and negative

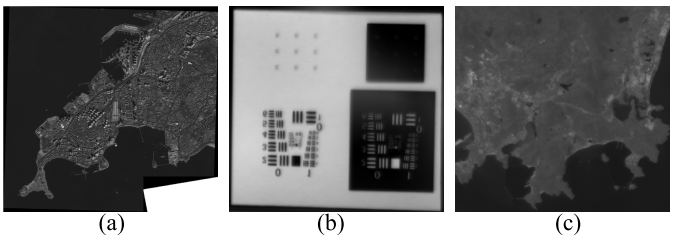


Fig. 1. Sample experimental data for verification. (a) Simulated data, (b) laboratory test data, and (c) GF-4 satellite data.

second-order differentials of in each iteration of signals. The algorithm solution of (12) is

$$\begin{aligned}\boldsymbol{w}^{k+1} = \underset{\boldsymbol{w}}{\operatorname{argmin}} \frac{\rho}{2}\left|\left|\boldsymbol{w} - \left(\nabla^2 \boldsymbol{z}^{k+1} + \frac{\mu_2^k}{\rho}\right)\right|\right|_2^2 \\ + \sum\nolimits_i \frac{\omega p |\boldsymbol{w}_i|}{\left(|\boldsymbol{w}_i^k| + \zeta\right)^{1-p}}.\end{aligned} \tag{13}$$

Let $\boldsymbol{m}_i = (\omega p / (|\boldsymbol{w}_i^k| + \zeta)^{1-p})$, where variable ζ is a small number, which is used to avoid dividing by zero. The solution of $\boldsymbol{w}$ can be obtained by the one-dimensional shrinkage function [26] as follows:

$$\begin{aligned}\boldsymbol{w}^{k+1} &= s\left(\nabla^2 \boldsymbol{z}^{k+1} + \frac{\mu_2^k}{\rho}, \frac{\boldsymbol{m}_i \omega}{\rho}\right) \\ &= \max\left\{\left|\nabla^2 \boldsymbol{z}^{k+1} + \frac{\mu_2^k}{\rho}\right| - \frac{\boldsymbol{m}_i \omega}{\rho}, 0\right\} \\ &\quad \cdot \operatorname{sign}\left(\nabla^2 \boldsymbol{z}^{k+1} + \frac{\mu_2^k}{\rho}\right)\end{aligned} \tag{14}$$

where $s(\cdot)$ denotes the one-dimensional shrinkage operator. $\operatorname{sign}(\cdot)$ is the sign function.

Lagrange multipliers μ_1 and μ_2 can be updated by the following formulae:

$$\mu_1^{k+1} = \mu_1^k + \rho\left(\boldsymbol{v}^{k+1} - \nabla \boldsymbol{z}^{k+1}\right) \tag{15}$$

$$\mu_2^{k+1} = \mu_2^k + \rho\left(\nabla^2 \boldsymbol{z}^{k+1} - \boldsymbol{w}^{k+1}\right). \tag{16}$$

The detail of the proposed MTFC algorithm is explained in Algorithm 1.

IV. Experimental Results and Analysis

A. Experiment Data

Three sets of data for experimental verification were used to quantitatively evaluate the performance of the MS-HONCTV-aided MTFC. The first set of data is the simulated degradation data. We selected a scene from the Worldview-3 satellite data acquired on October 16, 2014, in Qingdao, China. The spatial resolution of the data is 0.3 m. Next, we simulated the CMOS satellite acquisition of spaceborne surface array data and used random subpixel displacement mode to simulate image degradation. We degraded a frame of WorldView-3 satellite data into 24 LR images with random subpixel dislocation. After degradation, the spatial resolution of the LR images is 5 m. Since no suitable edge region could be found in the first set of simulated degradation data to test the MTF value, a point spread function degraded knife-edge target was added

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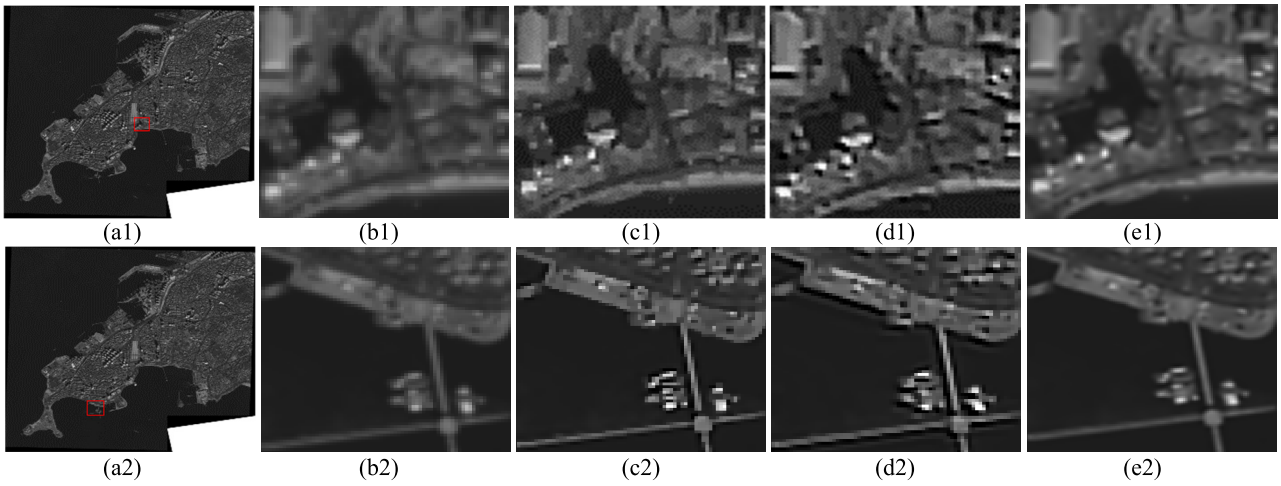


Fig. 2. Visual comparison of methods applied to simulated data. (a1) Original image, (b1) LR, (c1) ℓ_0 sparse representation, (d1) LMIP, and (e1) MS-HONCTV MTFC. (a2) Original image, (b2) LR, (c2) ℓ_0 sparse representation, (d2) LMIP, and (e2) MS-HONCTV MTFC.

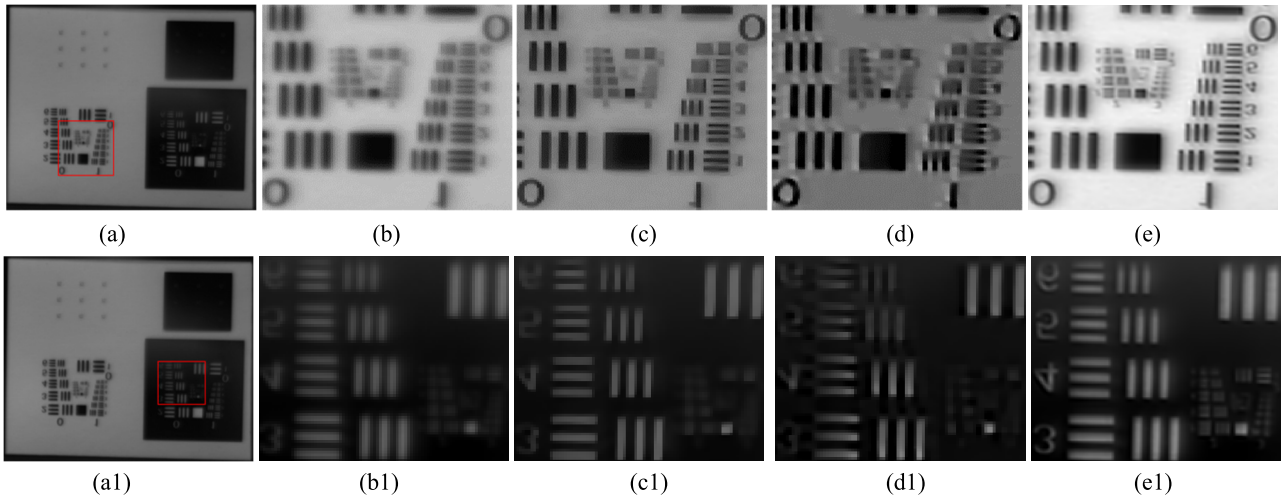


Fig. 3. Visual comparison of methods applied to laboratory data. (a) Original image, (b) LR, (c) ℓ_0 sparse representation, (d) LMIP, and (e) MS-HONCTV MTFC. (a1) Original image, (b1) LR, (c1) ℓ_0 sparse representation, (d1) LMIP, and (e1) MS-HONCTV MTFC.

to the bottom right corner of the original image, as shown in the white area in Fig. 1(a).

The second set of data was obtained from an optical simulation experiment in the laboratory. A motorized translation platform was used to translate an optical resolution target up and down from left to right. Eight images were acquired by continuous exposure of the area array to a CMOS camera, as indicated in Fig. 1(b). The shift interval of the translation platform was set 2 μm.

The third set of data consists of data from the GF-4 satellite. The spatial resolution of these data is 50 m. The data was acquired by the area array camera in staring imaging mode, which produces repeated observations of the target area in a short time (in the order of minutes) at high-frequency, which satisfies the criterion of HTD. The GF-4 satellite data set covers Sanya city and Hainan province, China. The data were captured on August 20, 2016, and the geographic coordinates are 18°09′ 34″ –18°37′ 27″ north latitude and 108°56′ 30″ –109°48′ 28″ east longitude. Since the width of a single-field GF-4 satellite image is 10 240 × 10 240 pixels, we cropped the same region of interest in the GF4 LR sequence image to determine the efficiency of the algorithm. The size of the cropped area is 512 × 1024 pixels. The present work uses eight consecutive frames of first-band data [see Fig. 1(c)].

B. Experiment Evaluation Metrics

The knife-edge method [27], [28] was used to evaluate the MTF in the experiments. The principle of the knife-edge method is to extract the edge spread function (ESF) from the image edge texture obtained from the optical imaging system. This is done by extracting the ESF of the image by fitting the gray distribution of the image edge. The derivative of the ESF is calculated to obtain the line spread function (LSF), and then the LSF is passed through Fourier transformation to obtain the MTF of the imaging system.

To further verify the improvement in image quality upon MTFC, the SNR [29], GMG, and SD were used to evaluate the performance.

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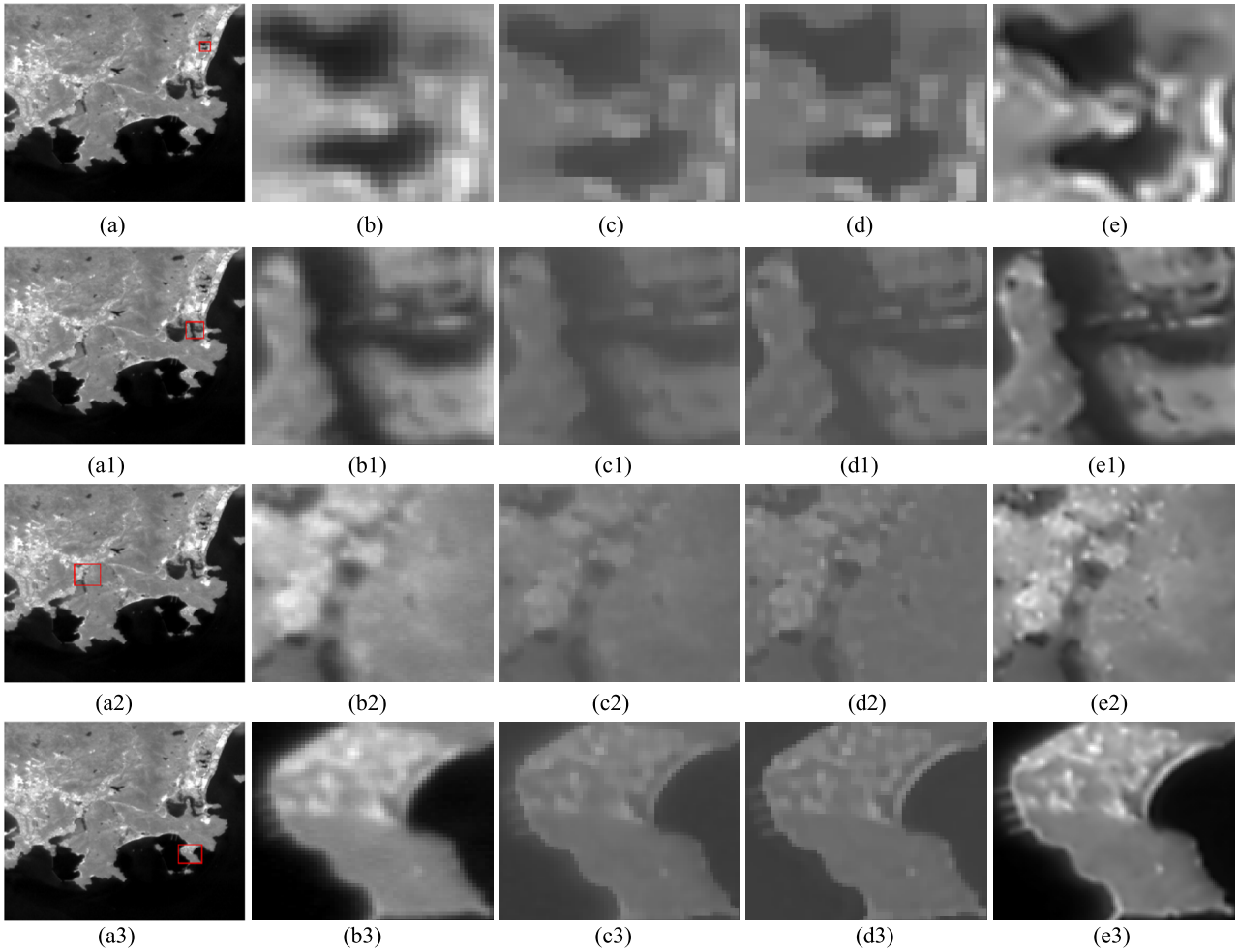


Fig. 4. Visual comparison of methods applied to GF4 satellite data. (a) Original image, (b) LR, (c) ℓ_0 sparse representation, (d) LMIP, and (e) MS-HONCTV MTFC. (a1) Original image, (b1) LR, (c1) ℓ_0 sparse representation, (d1) LMIP, and (e1) MS-HONCTV MTFC. (a2) Original image, (b2) LR, (c2) ℓ_0 sparse representation,(d2) LMIP, and (e2) MS-HONCTV MTFC. (a3) Original image, (b3) LR, (c3) ℓ_0 sparse representation, (d3) LMIP, and (e3) MS-HONCTV MTFC.

GMG is defined as

$$s(i', j') = \sqrt{\left(G_x(i', j')^2 + G_y(i', j')^2\right)} \tag{17}$$

where $s(i', j')$ is the GMG value with pixel coordinates of (i', j'). G_x and G_y represent the image gradients in x and y directions obtained by the Sobel gradient operator, respectively. The larger the GMG value is, the richer the image information and the better the algorithm compensation result.

SNR is defined by

$$\text{SNR} = \frac{\text{Signal}}{\sigma_s} \tag{18}$$

where Signal represents the signal magnitude and σ_s represents the image noise.

C. Experimental Verification

The proposed MS-HONCTV MTFC method is compared with other methods for the simulated data, the laboratory data, and the GF-4 satellite data. Considering that few MTFC methods are based on HTD, we compare our proposed method against two state-of-the-art single-frame deblur methods: ℓ_0 sparse representation image deblurring method [30] and the LMIP image deblurring method [15] as two benchmarks. All comparisons are made under the same conditions.

The visual results of simulation data and laboratory data are shown in Figs. 2 and 3, respectively. Compared with other methods, the proposed method exhibits clearer details in texture regions or edges, meanwhile produces fidelity results in smooth regions. A real GF4 satellite data MTFC result is shown in Fig. 4, further demonstrating the robustness of our method. It is shown that the proposed method can better remove image noise, preserve detailed information, restore smooth areas, and effectively eliminate staircase artifacts. Both the ℓ_0 sparse representation method and the LMIP method suppress blocky artifacts to some extent but over-sharpening the edges of the images.

The overall quantitative assessment results are shown in Table II where the performance metrics, GMG, SD, and SNR, obtained by all methods on simulation data, laboratory data, and GF-4 satellite data are collected. Obviously, the proposed MS-HONCTV MTFC method achieves the best performance reflected by the best SNR, GMG, and SD values.

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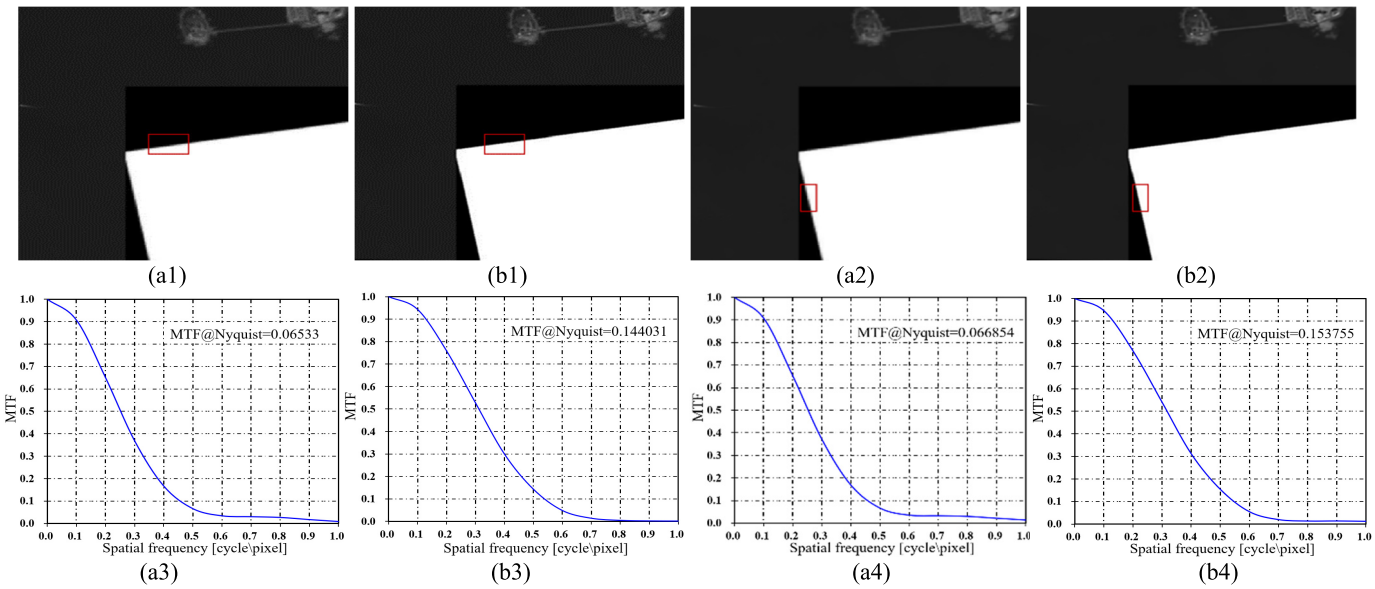


Fig. 5. Original data and result of MTFC on simulated data and corresponding MTF curves for the edge regions in red boxes. (a1) Original image and (b1) MTFC result. (a2) Original image and (b2) MTFC result. (a3) MTF curve of original image and (b3) MTF curve of MTFC result. (a4) MTF curve of original image and (b4) MTF curve of MTFC result.

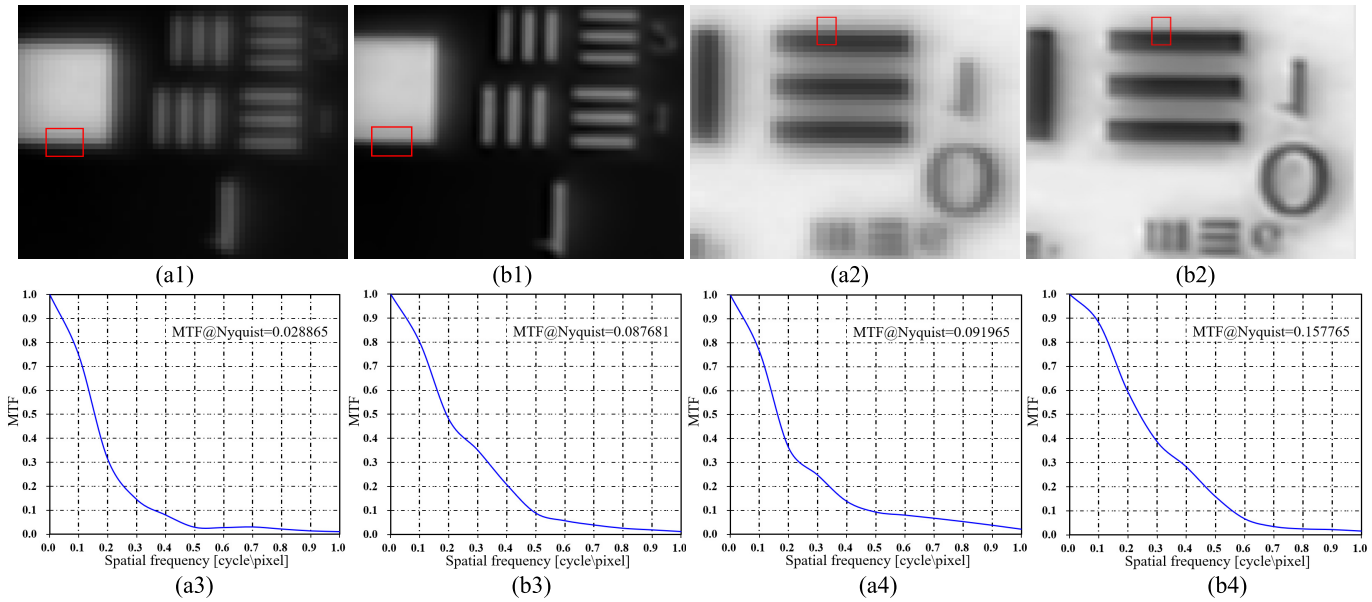


Fig. 6. Original data and MTFC results on laboratory data and corresponding MTF curves for the edge regions in red boxes. (a1) Original image and (b1) MTFC result. (a2) Original image and (b2) MTFC result. (a3) MTF curve of original image and (b3) MTF curve of MTFC result. (a4) MTF curve of original image and (b4) MTF curve of MTFC result.

Knife-edge method is used to evaluate MTF performance of three groups of experimental data results. Figs. 5–7 show that, after applying MTFC, the MTF curves for three groups of data are all higher than those for the original data and a higher MTF curve corresponds to better image quality. In addition, after applying MTFC at Nyquist cutoff frequency to the horizontal and vertical edge regions of the simulation data, the MTF values are better than those of the original images.

The MTF values for the horizontal edge region at the Nyquist cut-off frequency increase from 0.065 33 to 0.144 031, up by 120.5%, and the MTF values for the vertical edge region at the Nyquist cut-off frequency increase from 0.066 854 to 0.153 755, up by 130.0%. Similarly for laboratory data, the MTF values at the Nyquist cut-off frequency increased from 0.028 865 to 0.087 681 and from 0.091 965 to 0.157 765, respectively, for horizontal and vertical groups. The MTF values for GF4 satellite data at the Nyquist cut-off frequency increased from 0.028 89 to 0.089 797 and from 0.070 818 to 0.128 567. All test datasets show that at least 80% of improvement can be achieved with the proposed MTFC method.

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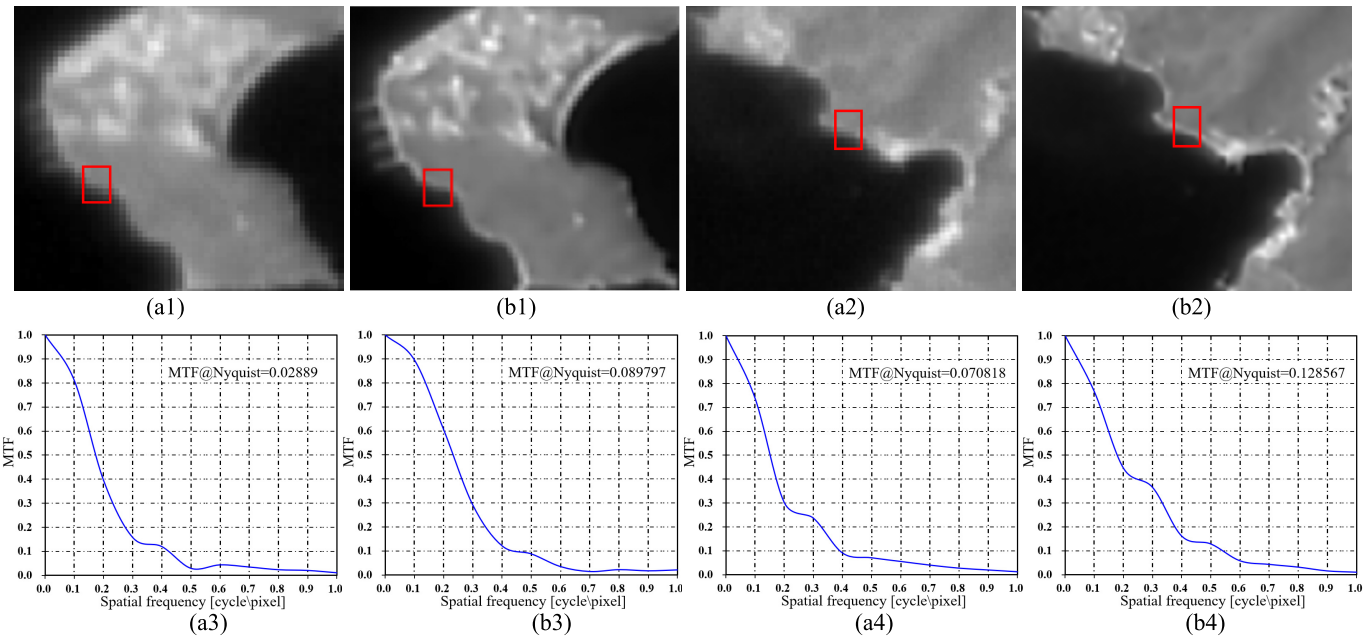


Fig. 7. Original data and MTFC results for GF-4 satellite data and corresponding MTF curves for the edge regions in red boxes. (a1) Original image and (b1) MTFC result. (a2) Original image and (b2) MTFC result. (a3) MTF curve of original image and (b3) MTF curve of MTFC result. (a4) MTF curve of original image and (b4) MTF curve of MTFC result.

TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT METHODS

Experiment	Method	SNR	GMG	SD
Simulation	Original simulated data	26.3225	0.0061	52.567
	ℓ_0 sparse representation	29.7269	0.0068	53.4469
	LMIP	35.7491	0.0074	54.5021
	MS-HONCTV MTFC	**44.1844**	**0.0076**	**55.1737**
Laboratory data	Original Laboratory test data	36.6883	0.0152	64.1071
	ℓ_0 sparse representation	39.4018	0.0163	65.2930
	LMIP	42.5257	0.0166	68.7783
	MS-HONCTV MTFC results	**50.8148**	**0.0167**	**72.773**
GF-4 data	Original GF-4 satellite data	54.4043	0.0056	81.2624
	ℓ_0 sparse representation	57.8202	0.0064	92.5925
	LMIP	62.8166	0.0068	92.8407
	MS-HONCTV MTFC results	**66.5593**	**0.0078**	**93.4260**

D. Analysis and Discussion

In the process of three data experiments on the MS-HONCTV MTFC method, several parameters must be carefully adjusted. In the experimental test, the blur kernel $c = 9 \times 9$, the total iteration number of the algorithm $\varrho = 30$; the warping matrix M_i and blurring matrix B_i in (4) are solved as described previously.

In each iteration of the MS-HONCTV MTFC method, the group sparse denoising problem is involved as defined in (10). Hence, the group size K have to be specified, which are set empirically. We found that $K = 3$ is a reasonable choice as the results after MTF compensation across three datasets are satisfying.

Because of the nonconvexity of the ℓ_p norm in (12). We investigated the most suitable value for p where $0 < p \leq 1$. We recommend $p = 0.5$ for practical use. The regularization parameter of the data fidelity term λ severely affects in eliminating the additive Gaussian noise. We set $0.5 < \lambda < 3$ and $1 < \omega < 3$ according to the noise level, blur kernel, and texture detail information of the three datasets.

Image blurring and noise greatly affect image quality, such as edges and structure. The proposed method is combined with the higher order nonconvex regularizer and group sparsity regularizer so that it can remove various noise effects, accurately restore edge structures, and can effectively eliminate staircase artifacts. Numerical results show that the proposed method

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Algorithm 1 MTFC Algorithm for HTD Based on MS-HONCTV Model

(1) Initialization:
  (a) Input parameters: $\boldsymbol{g}_{i=1,2,3\ldots N}$; blur kernel c;
  (b) set parameters: $\lambda > 0$, $\omega > 0$, $\rho > 0$, $k > 0$; the group size K, p;
  (c) the number of total iterations ϱ;
  (d) $\boldsymbol{z}^{\boldsymbol{0}} = 0$;
(2) Calculate the warping matrix M_i;
(3) Estimate the blur matrix for B_i;
(4) Begin the outer iteration loop index $\varrho = 1 : 2 : N$;
  (a) Calculate the residual $\mathcal{R}_e = \sum_{i=1}^{N} \left\{B_i M_i \boldsymbol{z}^{\varrho-1} - \boldsymbol{g}_i\right\}$;
  (b) Calculate the gradient $D_r = \sum_{i=1}^{N} \{B_i' M_i' R_e\}$;
  (c) Within inner iteration loop index $k = 1$;
    1) Update $\boldsymbol{v}^{k+1}$ by Eq. (10).
    2) Update $\boldsymbol{w}^{k+1}$ by Eq. (12).
    3) Update $\boldsymbol{z}^{k+1}$ by Eq. (8).
    4) Updated Lagrange multiplier μ_1^{k+1} by Eq. (15).
    5) Updated Lagrange multiplier μ_2^{k+1} by Eq. (16).
    6) if $\frac{||\boldsymbol{z}^k - \boldsymbol{z}^{k+1}||_2}{||\boldsymbol{z}^k||_2} \leq T$, (threshold $T = 1 \times 10^{-5}$) return
    7) end if
    8) $k = k + 1$
    9) end
  (d) Update $\boldsymbol{z}^{\varrho-1}$;
  (e) Calculate $\boldsymbol{z}^{\varrho-1} = \boldsymbol{z}^{\varrho-1} - D_r$;
  (f) end
(5) Quit the MTFC of image $\boldsymbol{z}^{\varrho}$.

increases SNR by 30%, GMG by 14.21%, and SD by 6.3% on average compared with the other state-of-the-art methods.

V. Conclusion

This article proposes the MS-HONVTV MTFC method. First, we analyzed the factors that affect the MTF of RS images. The analysis demonstrates that HTD based MTFC methods can theoretically improve the image quality produced by improving the detector's MTF. Second, an MTF degradation model and an effective MTFC total variation model were established by taking advantage of the higher order nonconvex regularizer and group sparsity regularizer. The IRL_1 method and the MM algorithm were employed in the ADMM framework to obtain the optimal solution of the HONCTV model. Experimental results showed that the MTFC of HTD improves MTF values with excellent image quality. Compared with the original images, the proposed MS-HONCTV method shows a significant improvement in terms of subjective visual evaluation, SNR, GMG, SD, and MTF output at Nyquist cut-off frequency. According to these results, the MS-HONCTV method holds the potential for GEORS imagery for quality enhancement, although the computational cost is not negligible. In future work, GPU will be considered to speed up the operation of the algorithm.

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